

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

23CSE304 Embedded Systems Laboratory

Activity Sheet

Activity Name: Seven Segment Display Interfacing

Name of Student:	Agneay B Nair
Registration No.:	CH.SC.U4CSE24102
Activity Date:	28.7.26
Submission Date:	28.7.26

Assessment Rubrics:

Component	Maximum Marks	Marks Allotted
Program	3	
Output	4	
Viva	2	
Timely Submission	1	
Total	10	

Faculty In-charge

Write an embedded C program for ARM architecture to interface Seven Segment Display. Debug and verify the program using IDE on a development computer system.

CODE:

```
#include<stdio.h>
#include<LPC213x.h>

unsigned char digit[10] = {
    0x3F, // 0
    0x06, // 1
    0x5B, // 2
    0x4F, // 3
    0x66, // 4
    0x6D, // 5
    0x7D, // 6
    0x07, // 7
    0x7F, // 8
    0x6F, // 9
};

void delay(int n){
    int i,j;
    for(i = 0;i<n;i++)
        for(j=0;j<1000;j++);
}

int main(){
    int i;
    PINSEL0 = 0x00000000;
    IODIR0 |= 0xFF;

    while(1){
        for(int i = 0;i<10;i++){
            IOCLR0 = 0xFF; // Clear Previous digit
            IOSET0 = digit[i]; // Display digit

            delay(500);
        }
    }
}
```

```
1  #include<stdio.h>
2  #include<LPC213x.h>
3
4  unsigned char digit[10] = {
5      0x3F, // 0
6      0x06, // 1
7      0x5B, // 2
8      0x4F, // 3
9      0x66, // 4
10     0x6D, // 5
11     0x7D, // 6
12     0x07, // 7
13     0x7F, // 8
14     0x6F, // 9
15 };
16
17 void delay(int n){
18     int i,j;
19     for(i = 0;i<n;i++){
20         for(j=0;j<1000;j++);
21     }
22 }
23
24 int main(){
25     int i;
26     PINSEL0 = 0x00000000;
27     IODIR0 |= 0xFF;
28
29     while(1){
30         for(int i = 0;i<10;i++){
31             IOCLR0 = 0xFF; // Clear Previous digit
32             IOSET0 = digit[i]; // Display digit
33
34             delay(500);
35         }
36     }
```

OUTPUT:



Registers

Regis...	Value
C...	0x40000070
	0x40000070
	0x40000070
	0x40000070
	0x00000000
	0x4000000C
	0x00000000
	0x00000000
	0x00000000
	0x00000000
	0x00000278
	0x00000000
	0x000001F0
	0x40000470
	0x000000E8
	0x000001F0
	0x600000D3
	0x00000000

Disassembly

```

26:      IODIR0 |= 0xFF;
27:
x000001FC E2410901 SUB      R0,R1,#0x00004000
x00000200 F5400008 LDR      R0,RO,#0x00000001

```

Third.c

```

19:   for(i = 0; i < n; i++)
20:       for(j = 0; j < 1000; j++)
21:   }
22:
23: int main() {
24:     int i;
25:     PINSEL0 = 0x00000000;
26:     IODIR0 |= 0xFF;
27:
28:     while(1) {
29:         for(int i = 0; i < 10; i++) {
30:             IOCLR0 = 0xFF; // Clear Previous digit
31:             IOSET0 = digit[i]; // Display digit
32:
33:             delay(500);
34:         }
35:     }
36: }
37:

```

Command Call Stack + Locals

